Civics Group	Index Number	Name (use BLOCK LETTERS)	H1

ST. ANDREW'S JUNIOR COLLEGE
2013 Preliminary Examination

H1 BIOLOGY **8875/1**

Paper 1: Multiple Choice

Friday 20 September 2013 60 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

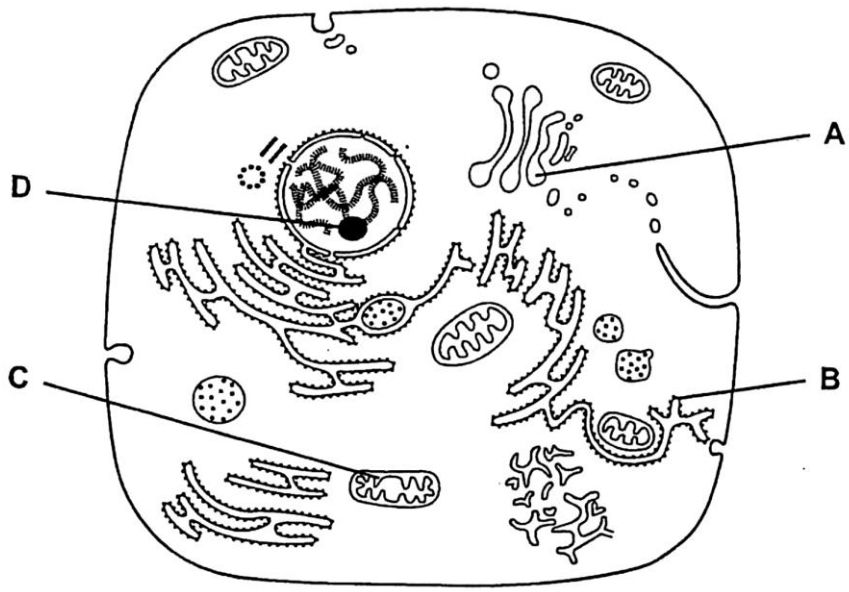
Each correct answer will score one mark. A mark will not be deducted for wrong answer. Any rough working should be done in this booklet.

At the end of the examination, submit the <u>multiple choice answer sheet</u> only.

This document consists of **14** printed pages.

[Turn over

1 The diagram below shows the ultrastructure of a eukaryotic cell. Which organelle does not contain nucleic acid?



2 A severe inherited condition arises from the failure to produce an enzyme that breaks down glycoproteins in cells. The condition can be diagnosed from an electron micrograph of a patient's cells. Which abnormality would be observed in these cells?

A An incomplete chromosome due to the lack of a gene.
B Larger lysosomes due to the accumulation of glycoproteins.
C Thinner cell surface membrane due to the lack of glycoproteins.
D Less endoplasmic reticulum due to a reduction in protein synthesis.

3 Which of the following comparisons between glycogen and cellulose is false?

	Glycogen	Cellulose
A	α-glucose residues are linked by $\alpha(1\rightarrow4)$ and $\alpha(1\rightarrow6)$ glycosidic bonds	β-glucose residues are linked by $\beta(1\rightarrow4)$glycosidic bonds
B	Branched	Unbranched
C	Made up of α-glucose monomers	Made up of amino acid monomers
D	Storage molecule	Structural molecule

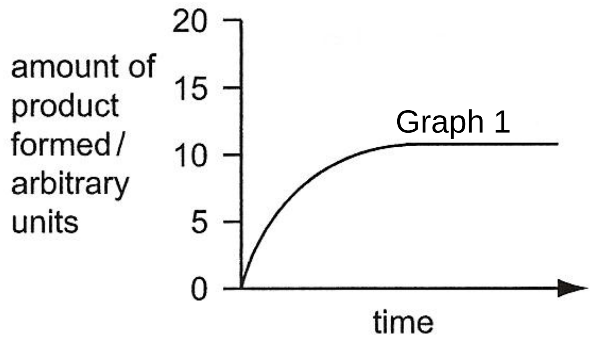
4 What does a haemoglobin molecule contain?

A four iron (Fe^{2+}) ions attached to each haem group
B four oxygen molecules attached to each haem group
C four polypeptide chains each with one attached haem group
D four polypeptide chains each with four attached haem groups

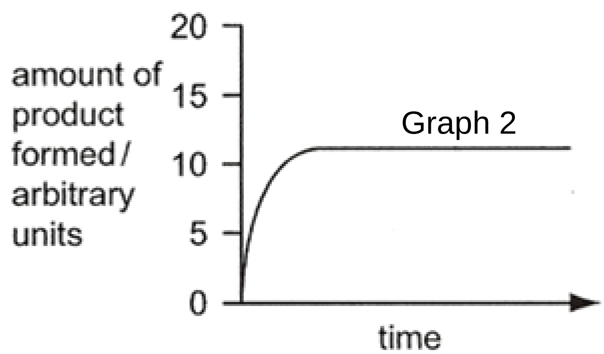
5 Which statement about lipids is correct?

A Phospholipids have hydrophobic regions to interact with cell cytoplasm unlike triglycerides.
B Phospholipids form a bilayer in double membranous organelles
C Triglycerides have a lower ratio of oxygen to carbon compared with carbohydrates.
D Triglycerides are made up of three fatty acids combined with glycogen.

6 Graph 1 below shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C and pH of 7.



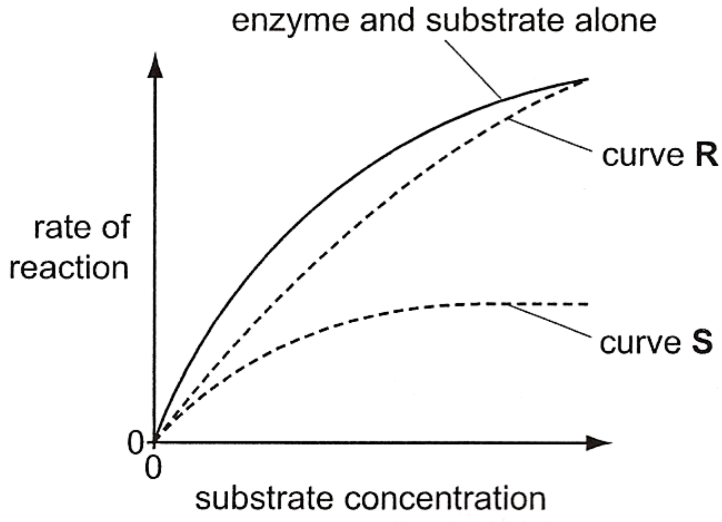
A change in condition(s) results in graph 2.



Which of the following conditions can best account for graph 2?

A Doubling of substrate concentration alone
B Doubling of enzyme concentration and increasing temperature to 20°C
C Increasing pH to 2
D Increasing temperature to 20°C alone

7 The graph shows the effect of two different compounds on the rate of reaction of the enzyme at different substrate concentrations.



Which statement is true?

A Compound which results in curve R increases the Km of the enzyme.
B Compound which results in curve R is structurally similar to enzyme active site.
C Compound which results in curve S decreases the Km of the enzyme.
D Compound which results in curve S does not change the 3D conformation of the active site.

8 A hypothetical diploid cell has four chromosomes and total amount of DNA in the cell is denoted by "X" before DNA replication.

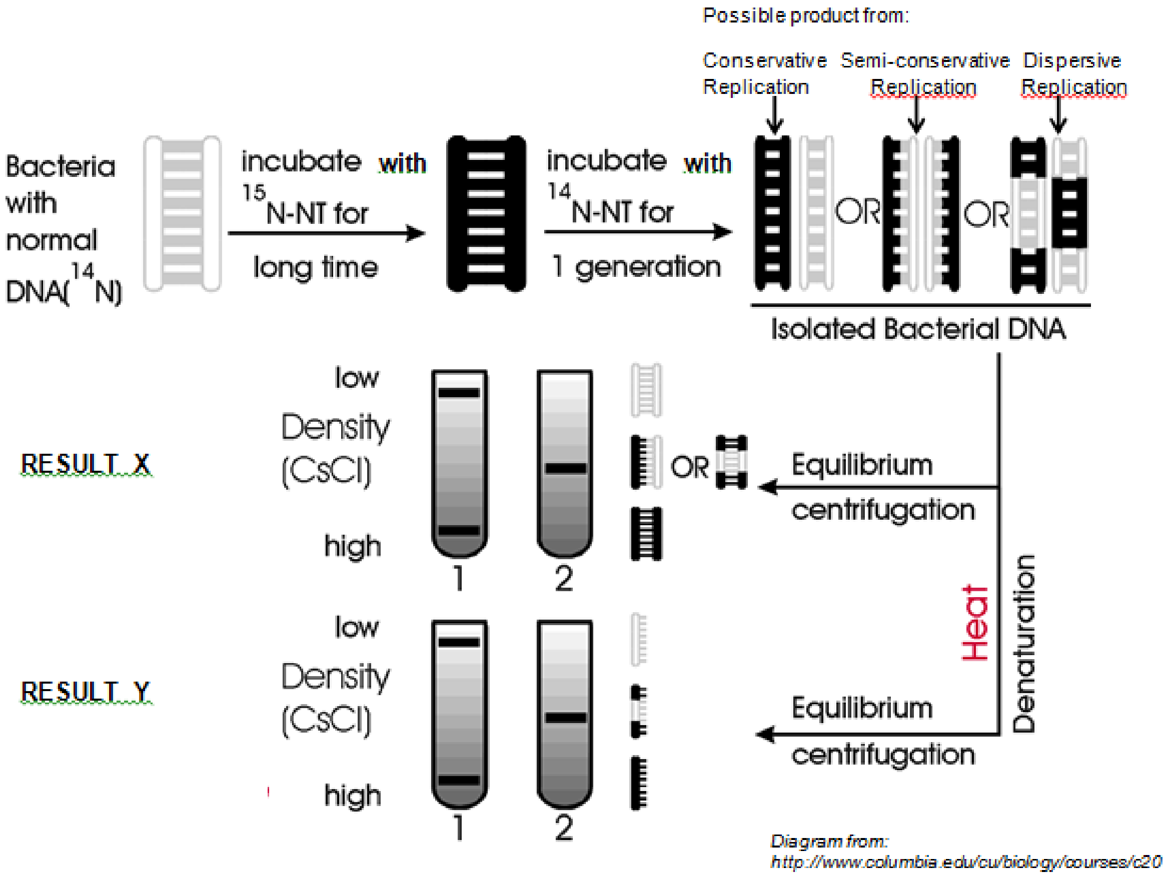
Which of the following is correct?

Option	Cell Cycle Phase	Number of chromosomes per cell	Amount of DNA per cell
A	Late Interphase	8	2X
B	Telophase of mitosis	8	X
C	Metaphase of Meiosis I	4	X
D	Anaphase of Meiosis II	4	X

9 When identical twins marry identical twins, the children of both couple are genetically

A identical unless crossing over takes place
B different because of random segregation during parental meiosis
C different because non-disjunction of chromatids occurs
D identical because of the effect of semi-conservative DNA replication prior to nuclear division

10 The Meselson-Stahl experiment is shown below. This experiment was done to prove that replication is via a semi-conservative mode rather than a conservative or dispersive mode.



Which result (tube 1 or 2) should be obtained, at each centrifugation X and Y, given that replication is semi-conservative? Which type of replication does the result obtained from centrifugation X and Y effectively eliminate?

	Centrifugation X		**Centrifugation Y**	
	Correct result	**Eliminates**	**Correct result**	**Eliminates**
A	1	Dispersive	1	Conservative
B	2	Conservative	2	Dispersive
C	1	Dispersive	2	Conservative
D	2	Conservative	1	Dispersive

11 To function as the heritable genetic code, DNA molecules must have all of the following structural features except

A the ability to form complementary base-pairs with RNA molecules.
B exist as a very stable, double helical form when being replicated
C a sequence of nucleotides that can be decoded into a sequence of amino acids in a protein
D ability to undergo semi-conservative replication

12 Which of the following order of steps is true for transcription in eukaryotes?

1 RNA polymerase II binds to promoter
2 Primase adds a RNA primer to the 3’ end of template strand
3 General transcription factors recognize and bind to TATA box of promoter
4 DNA polymerase III adds complementary deoxyribonucleotides to the 3’ end of the growing DNA chain
5 RNA polymerase II transcribes a DNA sequence which codes for a polyadenylation signal (AAUAAA) in the RNA transcript
6 RNA polymerase II transcribes a stop codon
7 RNA polymerase II adds complementary ribonucleotides to the 3’ end of the growing RNA chain
8 DNA polymerase III binds to promoter
9 Primer is hydrolyzed and replaced by deoxyribonucleotides

A **2, 1, 7, 5**
B **2, 8, 4, 9**
C **3, 1, 7, 5**
D **3, 1, 7, 6**

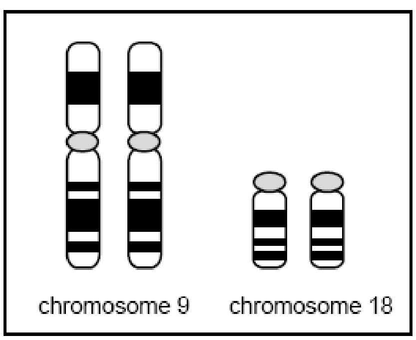
13 Translation did not occur successfully in a eukaryotic organism. Analysis of the structures in the cell revealed the following:

Structure	Presence/absence
Free floating aminoacyl-tRNAs	√
Initiator tRNA- small ribosomal subunit complex	√
Initiator tRNA (base-pairing with AUG codon) in the P-site of large ribosomal subunit	√
Second aminoacyl-tRNA in the A-site of large ribosomal subunit	√
Polypeptide chain	×

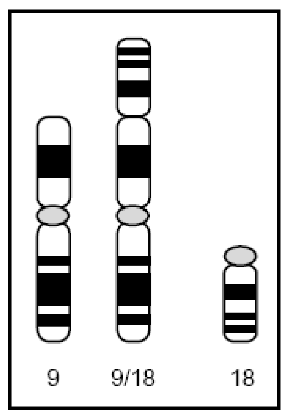
Which is the most probable explanation for these observations?

A aminoacyl tRNA synthetases fail to attach amino acids to tRNA
B there is an error in the scanning for the start codon by tRNA-small ribosomal subunit complex
C peptidyl-transferase is not functioning properly
D release factor fails to hydrolyze polypeptide chain from the tRNA

14 Genetic 'accidents' can occur to chromosomes, and are commonly known as chromosomal mutations. A scientist observed a chromosomal mutation involving chromosomes 9 and 18 in somatic cells of a male cat. The figure below shows chromosomes 9 and 18 in a normal male cat and in a cat with the chromosomal mutation.



Normal male cat



Cat with chromosomal mutation

The chromosomal mutation involved here is

A deletion.
B duplication.
C inversion.
D translocation.

15 Vitamin D resistant rickets is a sex-linked disease which results in impaired calcium absorption in the body. All female offspring of diseased males will get the disease.

A man without rickets marries a woman with rickets. The woman's mother did not have rickets. If this couple has a daughter, what is the probability that she will get rickets?

A 0
B 0.25
C 0.50
D 0.75

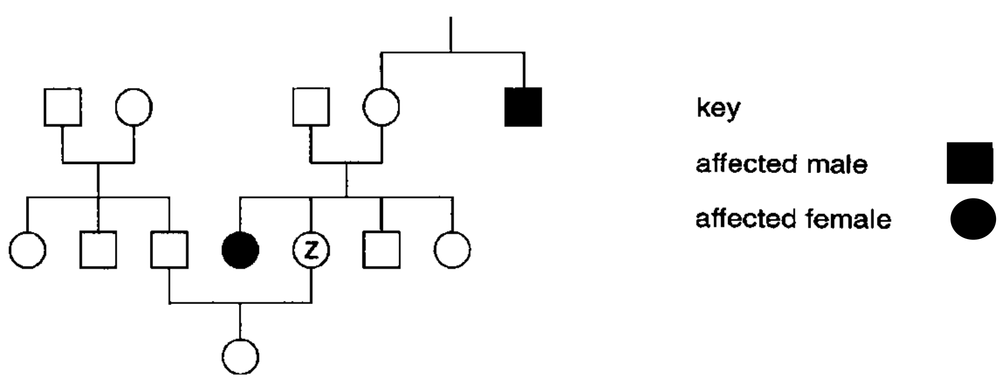
16 Two genes, Q and R affect the petals of a flower.

Gene Q has two alleles, Q^L and Q^A. The genotype $Q^L Q^L$ produces large petals, $Q^L Q^A$ produces small petals and $Q^A Q^A$ produces no petals.

Gene R has two alleles. R produces red pigment, and dominant over r that produces no pigment. Two plants, heterozygous for both genes are crossed. How many phenotypes are expected in the next generation?

A 4
B 5
C 6
D 7

17 Galactosaemia is a condition caused by a recessive allele of an autosomal gene in which galactose cannot be metabolised. The diagram shows the pattern of inheritance of galactosaemia in a family.



What is the chance that individual Z is homozygous dominant of the condition?

A 1/2
B 1/4
C 1/3
D 2/3

18 A mitochondria suspension obtained from liver cells is prepared for investigations of the products of respiration. Acetyl coA is added to the suspension.

Which of the following correctly matched the products of glycolysis and Krebs cycle for every oxidation of one glucose molecule in this mitochondria suspension?

	Product	Glycolysis	Krebs cycle
A	ATP	2	2
	NADH	2	6
	$FADH_2$	0	2
	CO_2	0	4

	Product	Glycolysis	Krebs cycle
B	ATP	0	2
	NADH	0	6
	$FADH_2$	0	2
	CO_2	0	4

	Product	Glycolysis	Krebs cycle
C	ATP	2	4
	NADH	2	4
	$FADH_2$	0	2
	CO_2	0	2

	Product	Glycolysis	Krebs cycle
D	ATP	0	4
	NADH	0	6
	$FADH_2$	0	2
	CO_2	0	4

19 What would be the effect of inhibition of lactate dehydrogenase in a mammalian cell under anaerobic conditions?

A A decrease in glycolysis, due to the lack of NAD^+.
B A decrease in cell pH, due to the accumulation of lactic acid.
C An increase in ATP production, due to increased amounts of reduced NAD.
D An increase in the activity of Krebs cycle, due to increased amounts of pyruvate.

20 Some photosynthetic organisms containing chloroplasts that lack PS II (photosystem II) are able to survive. The best way to detect the lack of PS II in these organisms would be to

A test for carbon fixation in the dark.
B test for liberation of oxygen in the light.
C determine the production of starch.
D determine if they have thylakoids in the chloroplasts.

21 Which of the following processes could still occur in a chloroplast in the presence of an inhibitor that prevents H^+ from passing through ATP synthase complexes?

1 Sugar synthesis
2 Photolysis of water
3 Transfer of electrons down the electron transport chain
4 Oxidation of NADPH

A 1 and 2
B 1and 4
C 2 and 3
D 3 and 4

22 The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A control experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

Which of the following is not consistent with the information in the table?

A At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
B At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
C In the dark, the concentration of carbon dioxide in the air after it had flowed over leave would be at least 280 ppm.
D At a light intensity of 10% of full sunlight, photosynthesis does not occur.

23 Which of the following statements were **not** part of Darwin's theory of evolution by natural selection?

1 Characteristics acquired during an organism's lifetime are passed to its offspring
2 Individuals in a sexually reproducing population are different.
3 Organisms produce more offspring than the environment can support.
4 Only those individuals best adapted to the environment survive and reproduce.
5 Regions that encode portions of the polypeptide that are vital for structure and function are less likely to incur mutations.

A 1 and 2
B 1 and 5
C 2 and 3
D 3 and 5

24 A comparison was made between human, rabbit, mouse and chimpanzee of the
• DNA coding sequence of the β globin gene
• DNA sequence in the introns of the β globin gene
• amino acid sequence of the β globin polypeptide

The data is shown in the table below.

	Sequence similarity (%)		
Organisms being compared	**Coding DNA**	**Intron**	**Amino acid sequence**
Human β globin /chimpanzee β globin	100	98.4	100
Human β globin /rabbit β globin	89.3	67	92.4
Human β globin / mouse β globin	82.1	61	80.1

It is possible to conclude from this data that

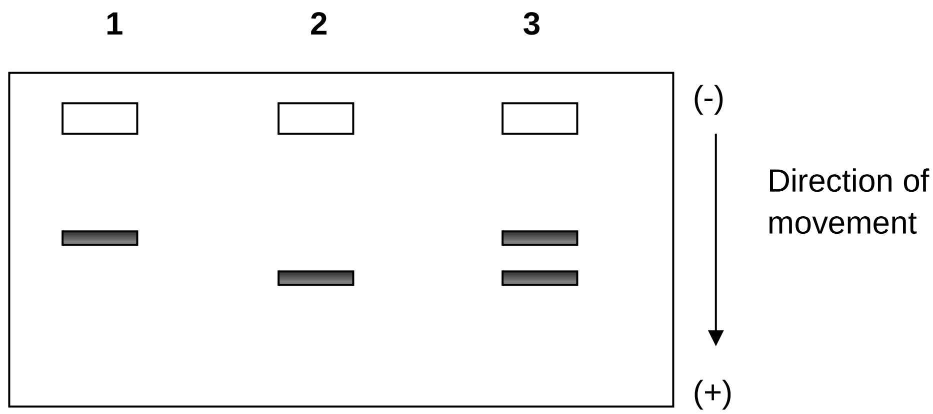
A a human is more closely related to a mouse than to a rabbit.
B the variation between chimpanzees and humans occurs in a region of the β globin gene which would code for amino acids.
C the variation in the intron sequence between human and mouse would account for some of the differences in the amino acid sequence.
D the comparison between rabbit and human indicates that the differences in their DNA did not always make a difference to the amino acid produced.

25 Which of the following descriptions refers accurately to homologous structures?

	Basic structure	Common ancestor	Function performed
A	Same	No	Different
B	Different	No	Same
C	Same	Yes	Different
D	Different	Yes	Same

26 The β-globin gene can exist in two different alleles termed HbA (the normal allele) and HbS (the allele that causes sickle cell anemia in homozygotes). The polypeptides that are coded for by these two alleles differ by one amino acid. Blood samples from 3 individuals were obtained and the **proteins** separated by gel electrophoresis.

Using the gel pattern below, determine the genotypes of individuals 1 and 2 and the reason for your identification.



	Genotype of 1	*Genotype of 2*	*Reason for identification*
A	Hb^AHb^A	Hb^SHb^S	Since glutamic acid in the normal β-globin is negatively charged, it will move faster towards the opposite pole.
B	Hb^SHb^S	Hb^AHb^A	Since valine in the normal β-globin is negatively charged, it will move faster towards the opposite pole.
C	Hb^AHb^A	Hb^SHb^S	Since glutamic acid in the β-globin that causes sickle cell is negatively charged, it will move slower towards the opposite pole.
D	Hb^SHb^S	Hb^AHb^A	Since valine in the β-globin that causes sickle cell is neutral, it will move slower towards the opposite pole.

27 Which of the following shows a correct comparison of the various processes?

	Polymerase Chain Reaction	**Transcription**
A	Occurs in the cytoplasm	Occurs in the nucleus
B	Requires nucleic acid	Does not require nucleic acids
C	Involves hydrogen bonding	Involves hydrogen bonding
D	Replicates the entire template strand	Replicates the entire template strand

28 Which of the following could be a future application of the technology, skills, and data acquired from the Human Genome Project?

1 Individuals can have their DNA sequenced.
2 Researchers can determine genetic variations among different populations of humans.
3 Drugs can be developed based upon an individual's genetic makeup.

A 1 only
B 2 only
C 1 and 2 only
D 1, 2 and 3

29 Which of the following statements are true about adult stem cells?

1 They can undergo self-renewal.
2 They can be totipotent, pluripotent or multipotent.
3 They can differentiate into almost any cell type.
4 They can give rise to specialised cells.

A 3 and 4
B 1 and 4
C 1, 2 and 4
D 1, 3 and 4

30 One type of genetically modified corn has

- A gene for the production of Bt toxin which protects the plant against a specific insect;
- A 'pat' gene for tolerance to the herbicide 'Basta'. This gene is used to select plants with the Bt toxin gene;
- An 'amp' gene which was introduced in the plant together with the Bt toxin gene. This gene gives resistance to the antibiotic ampicillin.

There is concern that the 'amp' gene may transfer to enterobacteria in the human intestine during nucleic acid digestion making treatment with ampicillin ineffective for disease caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

A An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.
B Bacteria cannot take up any DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.
C 50% of the enterobacteria isolated from humans are 'amp' resistant.
D All plant DNA is digested and destroyed in the human intestine during the digestion of plant cells by enzymes including human nuclease enzymes.